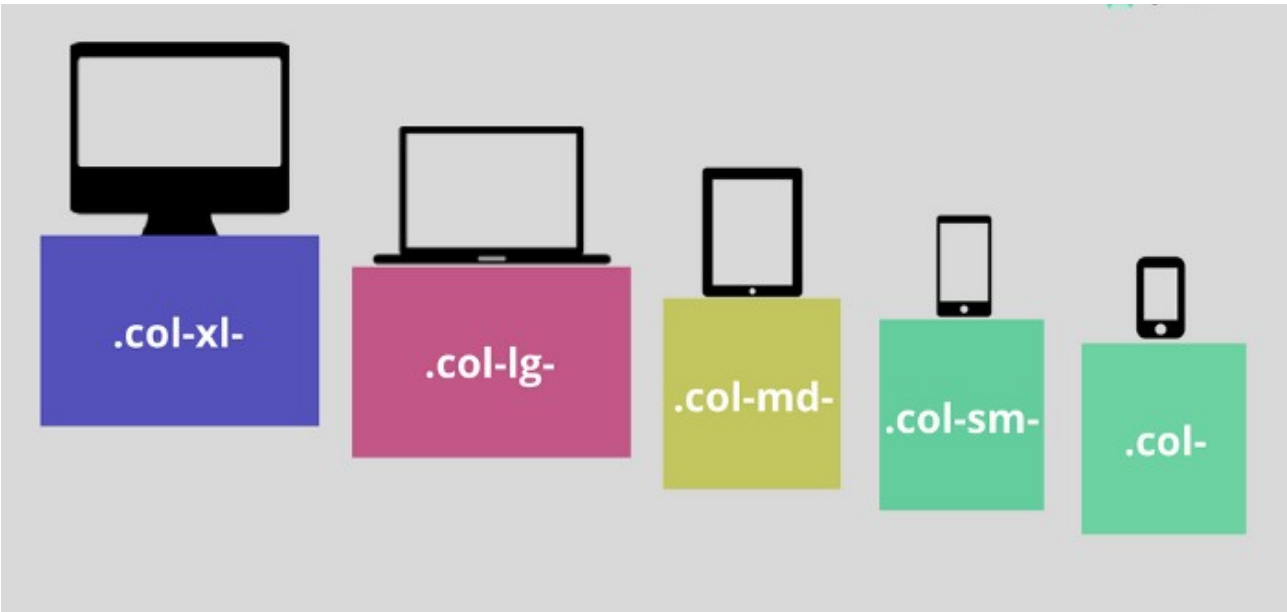
Bootstrap Grid System

What is Bootstrap Grid System?

Bootstrap grid system provides an easy and powerful way to create responsive layouts of all shapes and sizes. It is built with flexbox with mobile-first approach. Also, it is fully responsive and uses twelve column system (12 columns available per row) and six default responsive tiers.

You can use the Bootstrap's predefined grid classes for quickly making the layouts for different types of devices like mobile phones, tablets, laptops, desktops, and so on. For example, you can use the `.col-*` classes to create grid columns for extra small devices like mobile phones in portrait mode, and the `.col-sm-*` classes for mobile phones in landscape mode.

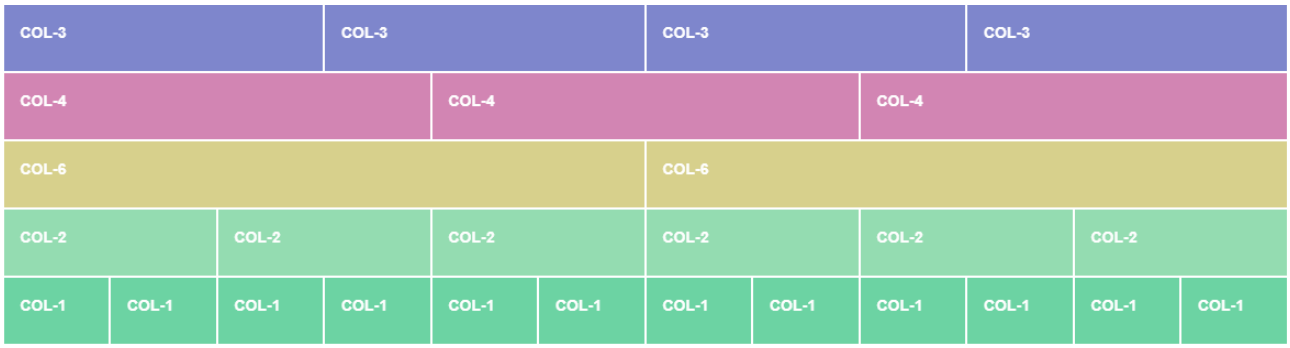
Similarly, you can use the `.col-md-*` classes to create grid columns for medium screen devices like tablets, the `.col-lg-*` classes for devices like small laptops, the `.col-xl-*` classes for laptops and desktops, and the `.col-xxl-*` classes for large desktop screens.



Bootstrap 5 Grid System

Bootstrap's grid system is built with flexbox and allows up to 12 columns across the page.

If you do not want to use all 12 columns individually, you can group the columns together to create wider columns:



Creating Two Column Layouts

The following example will show you how to create two column layouts for medium, large and extra large devices like tables, laptops and desktops etc. However, on mobile phones (screen width less than 768px), the columns will automatically become horizontal (2 rows, 1 column).

Example

```
<div class="container">
    <!--Row with two equal columns-->
    <div class="row">
        <div class="col-md-6">Column left</div>
        <div class="col-md-6">Column right</div>
    </div>

    <!--Row with two columns divided in 1:2 ratio-->
    <div class="row">
        <div class="col-md-4">Column left</div>
        <div class="col-md-8">Column right</div>
    </div>
```

```
    <!--Row with two columns divided in 1:3 ratio-->
    <div class="row">
        <div class="col-md-3">Column left</div>
        <div class="col-md-9">Column right</div>
    </div>
</div>
```

Creating Three Column Layouts

Similarly, you can create other layouts based on the above principle. For instance, the following example will typically create three column layouts for laptops and desktops screens. It also works in tablets in landscape mode if screen resolution is more than or equal to 992 pixels (e.g. Apple iPad). However, in portrait mode the grid columns will be horizontal as usual.

Example

```
<div class="container">
    <!--Row with three equal columns-->
    <div class="row">
        <div class="col-md-4">Column left</div>
        <div class="col-md-4">Column middle</div>
        <div class="col-md-4">Column right</div>
    </div>

    <!--Row with three columns divided in 1:4:1 ratio-->
    <div class="row">
        <div class="col-md-2">Column left</div>
        <div class="col-md-8">Column middle</div>
        <div class="col-md-2">Column right</div>
    </div>

    <!--Row with three columns divided unevenly-->
    <div class="row">
        <div class="col-md-3">Column left</div>
        <div class="col-md-7">Column middle</div>
        <div class="col-md-2">Column right</div>
    </div>
</div>
```

Additionally, you can also set the width of one column and let the sibling columns automatically resize around it equally. You may use the predefined grid classes or inline widths.

If you try the following example you'll find columns in a row with class `.col` has equal width.

Example

Bootstrap Auto-layout Columns

You can also create equal width columns for all devices (x-small, small, medium, large, x-large, and xx-large) through simply using the class `.col`, without specifying any column number.

Let's try out the following example to understand how it exactly works:

Example

```
<div class="container">
    <!--Row with two equal columns-->
    <div class="row">
        <div class="col">Column one</div>
        <div class="col">Column two</div>
    </div>

    <!--Row with three equal columns-->
    <div class="row">
        <div class="col">Column one</div>
        <div class="col">Column two</div>
        <div class="col">Column three</div>
    </div>
</div>
```

Additionally, you can also set the width of one column and let the sibling columns automatically resize around it equally. You may use the predefined grid classes or inline widths.

If you try the following example you'll find columns in a row with class `.col` has equal width.

Example

```
<div class="container">
    <!--Row with two equal columns-->
    <div class="row">
        <div class="col">Column one</div>
        <div class="col">Column two</div>
    </div>

    <!--Row with three columns divided in 1:2:1 ratio-->
    <div class="row">
        <div class="col">Column one</div>
        <div class="col-sm-6">Column two</div>
        <div class="col">Column three</div>
    </div>
</div>
```